الكلية الجامعية للعلوم التطبيقية
University College of Applied Sciences

Embedded Systems
COMP 4315

Instructor Info

Dr. Mohammed A. Matar

Office Hrs: Sat, & Wed 13:00-14:00

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Course Info

Prereq: Beginner Guides to Embedded Systems and Robotics, Sourabh Sankule, 2009.

Sat & Wen

08-09.30

W209

Course Description

Fundamentals of computer systems design, interfacing and basics of embedded computers (microprocessors). Laboratory projects require students to successfully design, implement, debug, and document computer solutions requiring a mix of hardware and software. Introduction of models and methodologies for designing systems containing both hardware and software components, and co-design. Substantial design projects will be required of each student.

Course Textbook

Required Texts

Designing Embedded Systems with PIC Microcontrollers - Principles and Applications, Tim Wilmshurst, 1st edition, 2007, Published by Elsevier.

Recommended Text

First Steps with Embedded Systems, by Byte Craft Limited, 5th edition.

Other

Any required journal articles and book chapters will be provided.

Grading Scheme

10%	Quizzes
10%	Homework
5%	Class activities
25%	Midterm exam
50%	Final term

Goals

First Part

Knowing the various generations of microprocessors, as well as the main parts of these microprocessors.

Last Part

learning basics of microprocessor interfacing with other main parts of computers.

Learning Outcomes

First Part

- The student knows the various versions and generations of Microprocessors, as well as the main manufacturers.
- The student knows cons and pros of versions covered in the textbook.
- The student is familiar with different addressing modes used in microprocessors, in addition to basics of both data movement and arithmetic & login instructions.
- The student can deal with main control instructions used for programming microprocessors.

FAQs

How difficult will the quizzes be?

They will not be that difficult, they will be a very good tool to review the chapters.

What is the nature of the required assignments?

They will be research topics related to the studies chapters.

Can we do some projects to have bonus points?

Yes, any good project related to the course will be evaluated and can be accounted as bonus points.

Last Part

- The student knows the architecture of 8088/8086 microprocessors.
- The student knows the block diagrams and general schematic of memory interface with microprocessors in computers.
- The student is familiar with basic input/output of computers and how they are interfaced to microprocessors.
- The student knows various types of interrupts and how microprocessors deal with them.
- The student knows how microprocessors access different memory parts using direct memory access mode, as well as Bus interface basics.

Quizzes

There will be ten to fifteen quizzes given throughout the semester, of which the highest ten will be used in your final grade. These will be to reward students who consistently show up to class, but will be more than just attendance points.

Homework

Homework is another example of learning by doing. Although not as exciting as a lab, homework is essential to learning the concepts in this course. Homework will be in the form of reading assignments and problem sets, with a due date 2-3 lectures after it is assigned. Homework must be done individually (you will learn the most from this). There will be 10-13 assignments, of which the highest 10 will be used in your final grade.

Exams

There will be one mid-semester exam and one final exam. Exams are closed-book, closed-notes. Mobiles phones are not permitted. Exams will include material from the lecture, the quizzes, and homework.

Course Lectures

We will use PowerPoint slides to teach this class. You can download them and print them from my web page. See the web for the course lecture outline.

General Notes

- The standards and requirements set forth in this syllabus may be modified at any time by the course instructor. Notice of such changes will be by announcement in class and/or by email to your UCAS email address.
- Please note that mobile phones, classroom chatter, and late arrivals are disruptive to the instructor and to your peers.
- If you have any question your are welcome to ask in the lecture, office hours or via email.

Class Schedule

FIRST PART

Week 1	Introduction To Embedded Systems	Embedded Systems basics and applications.
Week 2	Tiny computers, hidden control	The meaning of the term ‘embedded system’; The Microcontroller which lies at the heart of the embedded system; The Microchip PIC® family; A first PIC Microcontroller, the 12F508.
Week 3	Introducing the PIC® 16 Series and the 16F84A	The PIC mid-range family, in overview; The overall architecture of the 16F84A; The 16F84A memory, and a review of memory technologies; Other hardware features of the 16F84A, including reset.
Week 4	Parallel ports	Why we need parallel input/output; How simple parallel I/O logic circuits can be developed; How external devices can be connected to the parallel port; The parallel I/O available on the PIC 16F84A.
Week 5	Power supply and the clock oscillator	The essential features of power supply and clock oscillator; The Microchip approach to power supply and oscillator; The hardware design of the electronic ping-pong game.
Week 6	Starting to program an introduction to Assembler	The essentials of Assembler programming; Microchip MPLABTM Integrated Development Environment; PIC 16 Series instruction set, and use of certain instructions; Program simulation, and the MPLAB simulator MPSIMTM; Program download to a target system.
Week 7	Larger systems and the PIC® 16F873A	The architecture of the 16F873A Microcontroller; The 16F873A memory map; The 16F873A interrupt structure; The use of the Microchip in-circuit debugger.
Week 8	Review	First Part
	EXAM	MIDTERM

LAST PART

Week 9	The human and physical interfaces	Human interfacing needs and some simple means of meeting these; Some simple example sensors; Some ways of interfacing between sensor signals and the Microcontroller; Some simple example actuators; Some ways of interfacing between the Microcontroller and the actuator.
Week 10	Taking timing further	Further applications of counting; Recording ('capturing') in timer hardware the time an event occurs; Using timer hardware to trigger events at particular times; Generating repetitive time-based events, including PWM; Measuring frequency.
Week 11	Starting with serial	The principles of synchronous serial communication; the implementation of the Serial Peripheral Interface (SPI); the implementation of the I2C (Inter-Integrated Circuit) protocol; the principles of asynchronous serial communication; the implementation of asynchronous serial communication.
Week 12	Data acquisition	The main features of a data acquisition system; the characteristics of an analog-to-digital converter; the characteristics of the 16F873A analog-to-digital converter; how the 16F873A analog-to-digital converter can be applied.
Week 13	Data manipulation	Some simple data manipulation techniques; the use of comparators and the 16F873A comparator capability.
Week 14	Smarter systems and the PIC® 18FXX2	The architecture of the 18 Series Microcontroller, focusing on the 18F2420; The 18 Series instruction set; The memory structure, and how it is accessed and addressed.
Week 15	The PIC® 18FXX2 peripherals	The 18FXX20 peripherals, in most cases drawing on knowledge of their 16 Series equivalents.
Week 16	Review	Last Part
	EXAM	FINAL